A First Lean Formalization Audit of the IUT III Corollary 3.12 Corridor

IUT Lean formalization project

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Abstract

This note records the present state of a Lean formalization of a narrow part of Mochizuki's inter-universal Teichmuller theory: the passage from IUT III, Theorem 3.11 to Corollary 3.12, with attention to the Scholze–Stix objection. The current Lean code proves conditional route theorems, obstruction lemmas, and the final ordered-real algebra in Step (xi-f). It does not prove Corollary 3.12 and does not refute it. The main formal result so far is that a supplied hull/log-volume q-to-Θ comparison gives $C_\Theta \geq -1$, and that a nonarchimedean (Ind3) entry together with explicit Hodge/SHE square-weight transport data, or the primitive factored square/full-label SHE obligations, feeds such a comparison, while several simplified collapse mechanisms are rejected by Lean.

1 Scope

The review used the local Markdown conversions of IUT I–IV, Mochizuki's 2026 formalization note, and the Scholze–Stix note. The relevant source passages are:

- IUT I: initial Θ-data, prime strips, Hodge theaters, and non-scheme-theoretic Θ-links.
- IUT II: Hodge–Arakelov evaluation, theta values of shape q^{j^2}, multiradiality, and Gaussian monoids.
- IUT III: Theorem 3.11; Corollary 3.12, especially Steps (x) and (xi); Remarks 3.12.2 and 3.12.3.
- IUT IV: later explicit log-volume estimates and Diophantine consequences.
- Scholze–Stix: the ordered real-line identification problem in their Section 2.2.
- Mochizuki 2026: the "3.11.5 ⇒ 3.12" fourth-triangle decomposition.

2 Mathematical Target

IUT III Step (x) treats procession-normalized log-volumes as invariants for (Ind1) and (Ind2), and turns (Ind3) into an upper inequality. Step (xi) then uses the multiradial construction, IPL, SHE, holomorphic hulls, determinants, and log-volume to place the q-pilot log-volume in an upper ray controlled by the Θ-pilot log-volume:

$$-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|} \quad \Rightarrow \quad -|\log(q)| \leq -|\log(\Theta)|.$$

If C_Θ is any real number satisfying

$$-|\log(\Theta)| \leq C_\Theta |\log(q)|, \qquad |\log(q)| > 0,$$

then Step (xi-f) concludes $C_\Theta \geq -1$. This last inference is now a Lean theorem. Mochizuki's 2026 note isolates the final comparison as:

$$\text{Theorem 3.11} \xrightarrow{\text{APT+IPL}} \text{“3.11.5”} \xrightarrow{\text{SHE+IPL}} \text{Corollary 3.12.}$$

The present formalization targets only this corridor, not the full construction of Theorem 3.11.

3 Dispute Interface

Scholze–Stix require explicit bookkeeping for all ordered one-dimensional real vector spaces. Their pressure point is that the Θ-side concrete objects carry j^2-scaled degrees, while a consistent identification of the real lines appears to leave no room for inserting all these scalars.

The formalization therefore separates:

$$\begin{array}{c} \text{representative } j^2 \text{ data} \\ \text{balanced sign-compatible data} \\ \text{aggregate averaged data} \\ \text{final hull/log-volume data} \end{array} \quad \text{and} \quad \begin{array}{c} \text{raw real equality} \\ \text{transported equality} \\ \text{matched transport scales} \\ \text{cancellable ordered-line comparison.} \end{array}$$

This separation is the main correctness guard in the current code.

4 Lean Coverage

The relevant implementation is in

`Iut/Stage1/IUTStage1Source.lean` and `Iut/Stage1/IUTStage1Experiments.lean`.

The code currently covers:

- labeled real-line copies and positive-scale transports;
- a finite $\mathbb{F}_\ell$-label model with representative square weights $j \mapsto j^2$;
- a representative pilot-degree profile $\deg(\Theta_j) = j^2 \deg(q)$;
- proof that one label-independent scale cannot absorb all representative j^2 scales in this model;
- proof that, for nonzero $\deg(q)$, one label-independent scale cannot account for all representative Θ_j-degrees;
- proof that the raw representative j^2 theta-degree rule does not descend to the sign quotient $|F_\ell| = F_\ell/\{\pm 1\}$ when $\deg(q) \neq 0$;
- an absolute-label exponent on $|F_\ell| = \{0\} \cup F_\ell^\times/\{\pm 1\}$ which agrees with j^2 for the canonical representatives $0 \leq j \leq (\ell - 1)/2$, and the corresponding theta-degree profile; Lean proves that the coordinate map $F_\ell \to |F_\ell|$ is surjective, that the zero absolute label is reached exactly by $j = 0$, hence has singleton coordinate fiber, and that every nonzero sign-label class has a nonzero coordinate representative; Lean also proves the exact fiber relation: two

coordinates have the same full absolute label if and only if they are equal or negatives of one another; equivalently, every nonzero full-label fiber is exactly a sign pair $\{j, -j\}$, hence has cardinality 2. Lean also proves that the half-range procession map to $|F_\ell|$ is bijective, using the minimal absolute representative $j_{\min}$ for surjectivity and the exponent j^2 for injectivity; this gives $\#|F_\ell| = (\ell+1)/2$ in the finite model and permits Lean to reindex absolute-label averages by half-range processions; Lean also proves that the half-range procession top is exactly $(\ell-1)/2$ and that the resulting finite index set has cardinality $(\ell+1)/2$, matching the $j \in \{0, \ldots, (\ell-1)/2\}$ convention in IUT II; on this half-range procession, Lean proves the Gaussian degree formula $G(j) = j^2 \deg(q)$, with $G(0) = 0$, and the normalized average identity $6\ \mathrm{Avg}(G) = j_{\max}(2j_{\max}+1)\deg(q)$; by reindexing, the same identity holds for the average over $|F_\ell|$. Since $j_{\max} \geq 2$, Lean proves that this full average is not the singleton $j = 1$ identity value when $\deg(q) \neq 0$; in the nonpositive environment branch, the full $|F_\ell|$-average is bounded above by $\deg(q)$, and hence by any bound above $\deg(q)$; if $\deg(q) < 0$, this bound is strict: the full average is $< \deg(q)$, while for $\deg(q) > 0$ it is $> \deg(q)$; the same facts are now packaged as a typed $|F_\ell|$-indexed procession-normalized average object, where the average is provably distinct from the canonical $j = 1$ component unless $\deg(q) = 0$, strictly below it if $\deg(q) < 0$, strictly above it if $\deg(q) > 0$, and equal to that component if and only if $\deg(q) = 0$;

- the degree-level Gaussian evaluation $q \mapsto (q^{j^2})_{j \in |F_\ell|}$, including identity at $j = 1$; Lean now also converts such an evaluation into a cusp/full-label log-volume compatibility object and proves the full-label value-preservation branch when the source and target environment degrees agree; Lean also proves the IUT II restriction check at the canonical $j = 1$ sign label: the Gaussian component and the corresponding cusp-class log-volume are both the original environment degree; Lean also records the accompanying IUT II warning in the finite model: the singleton restriction to $j = 1$ is not stable under the additive F_ℓ-torsor action, since translation by 1 sends 1 to $2 \neq 1$; more generally, Lean proves that every nonempty translation-stable finite subset of F_ℓ is all of F_ℓ, so no nonempty proper subset restriction is compatible with this symmetry; Lean also proves that no nonzero additive translation descends to a well-defined map on $|F_\ell|$, since the equal absolute labels of t and $-t$ would have to map simultaneously to the nonzero label of $2t$ and to 0; equivalently, an additive translation descends to $|F_\ell|$ if and only if it is the zero translation; the same iff is exposed at the route interface as the exact condition for a translation equivalence to preserve full labels;

- a canonical representative square-weight profile on F_ℓ: Lean proves the total j^2-mass is positive from the $j = 1$ contribution and therefore constructs the profile without an external positivity premise; it also proves $6\sum_{j\in\mathbb{Z}/\ell\mathbb{Z}} j_{\mathrm{rep}}^2 = (\ell-1)\ell(2(\ell-1)+1)$, hence the corresponding average formula after division by ℓ;

- the square-weighted constant-average check: a constant procession-normalized log-volume remains equal to the same constant after averaging with the canonical j^2-weights;

- the same constant-average check inside the $F_\ell = \mathbb{Z}/\ell\mathbb{Z}$ cusp-label corridor: if the full-label log-volume is constant, the square-weighted average used in the Θ-route is that constant, including for the canonical weights;

- the constant-target structured-SHE subcase: the formal route now proves equality of the square-weighted average with the Θ-source average, not merely an upper bound;

- a log-volume shadow of the IUT III splitting-monoid action: the generator attached to a procession label contributes $j^2 \deg(q)$, contributes 0 at the core label, and adds its finite generator sum to the tensor-packet log-volume; Lean also proves the procession-normalized form: the acted tensor-packet average is the original tensor-packet average plus the averaged Gaussian generator contribution, equivalently the average of the $j^2 \deg(q)$ contributions over the procession container; for $S_{j+1} = \{0, \ldots, j\}$, Lean expands the total generator contribution to $j(j+1)(2j+1) \deg(q)/6$, and the resulting normalized log-volume change to $j(2j+1) \deg(q)/6$; since $j = \lfloor \ell/2 \rfloor \geq 2$, a nonnegative environment degree makes the acted normalized tensor-packet log-volume an upper bound for the original one;

- the finite procession containers $S^{\pm}_{j+1} = \{0, \ldots, j\}$, their nested inclusions, core-label stability, stage cardinality $j+1$, and total procession ambiguity $(\ell^{\pm})!$ rather than $(\ell^{\pm})^{\ell^{\pm}}$;

- the map from the final procession container to $|F_\ell|$, sending the core label to 0 and recovering the exponent j^2;

- the log-volume shadow of the tensor packet over $S^{\pm}_{j+1}$, with each factor a finite-fiber direct sum over places above $v_{\mathbb{Q}}$, total log-volume given by the finite sum over the container, normalized denominator $j+1$, and invariance under container permutations;

- the F_ℓ-label average specialization: for labels represented as $\mathbb{Z}/\ell\mathbb{Z}$, Lean rewrites the procession-normalized average denominator as ℓ;

- the zero/nonzero Gaussian average split: Lean proves

$$\#(\mathbb{Z}/\ell\mathbb{Z})^{\times} = \ell - 1, \qquad \mathrm{Avg}_{j \in F_\ell} G(j) = \frac{\ell - 1}{\ell} \mathrm{Avg}_{j \neq 0} G(j),$$

  for the degree-level Gaussian evaluation G, using $G(0) = 0$. Thus the paper's uniform $1/\ell$ weighted-volume convention is not silently replaced by the auxiliary nonzero-only average; in the negative signed branch Lean also proves the strict inequality $\mathrm{Avg}_{j \neq 0} G(j) < \mathrm{Avg}_{j \in F_\ell} G(j)$, while in the positive branch the inequality is reversed; equality between the two averages holds if and only if $\deg(q) = 0$. Lean also proves that the all-F_ℓ coordinate average itself is zero if and only if $\deg(q) = 0$; Lean separately formalizes the nonzero half-range $j = 1, \ldots, (\ell - 1)/2$ average, proves that the signed nonzero carrier $F_\ell^{\times}$ is the twofold cover of this half-range by j and $-j$, and therefore proves that the signed nonzero average equals the nonzero absolute-label average. It also proves $6\,\mathrm{Avg}_{j \neq 0,\, |F_\ell|} G(j) = (j_{\max} + 1)(2j_{\max} + 1) \deg(q)$; Lean also proves the exact rescaling

$$\mathrm{Avg}_{|F_\ell|} G = \frac{j_{\max}}{j_{\max} + 1} \mathrm{Avg}_{j \neq 0,\, |F_\ell|} G.$$

  Since the full $|F_\ell|$-average has coefficient $j_{\max}(2j_{\max} + 1)/6$, Lean proves that the nonzero absolute-label average is strictly below the full average when $\deg(q) < 0$ and strictly above it when $\deg(q) > 0$, with equality if and only if $\deg(q) = 0$. Combining the signed and absolute computations, Lean proves

$$\mathrm{Avg}_{F_\ell} G = \frac{j_{\max}(j_{\max} + 1)}{3} \deg(q),$$

  equivalently $\mathrm{Avg}_{F_\ell} G = ((\ell + 1)/\ell) \mathrm{Avg}_{|F_\ell|} G$, so $\mathrm{Avg}_{F_\ell} G$ is strictly below $\mathrm{Avg}_{|F_\ell|} G$ when $\deg(q) < 0$, strictly above it when $\deg(q) > 0$, and equal to it only when $\deg(q) = 0$;

- the IUT II Remark 4.7.3 zero/nonzero weighted-volume relation: Lean records a relation in which every nonzero full label is subordinate to the zero label, proves that $0 \ll 0$ is false, proves that this relation is not symmetric at the zero label, and proves the exact criterion $a \ll 0 \iff a \neq 0$ for full labels; at coordinate level this becomes $j \ll 0 \iff j \neq 0$, so the all-coordinate statement is false precisely because it includes $j = 0$. Lean also proves that the full-label set subordinate to zero has cardinality $(\ell - 1)/2$, the nonzero absolute half-range, and decomposes any full-label sum, hence any full-label average, into its zero contribution plus the subordinate nonzero contribution with denominator $(\ell + 1)/2$. Applying this to the Gaussian degree function, whose zero component is 0, Lean rewrites the full absolute-label Gaussian average as the subordinate nonzero Gaussian sum divided by the full absolute-label denominator and proves the closed form

$$6 \sum_{a \ll 0} G(a) = j_{\max}(j_{\max} + 1)(2j_{\max} + 1) \deg(q).$$

  Lean also proves that this subordinate sum is exactly the canonical nonzero absolute half-range sum $\sum_{j=1}^{j_{\max}} G(j)$. In contrast to nonzero additive translations, Lean defines the multiplicative unit action on full labels, proves its compatibility with the coordinate formula $j \mapsto a \cdot j$, and proves that it preserves the subordinate zero/nonzero relation. Lean also proves the unit and multiplication laws for this action and proves that the sign subgroup $\{\pm 1\}$ acts trivially on full absolute labels. Each unit action is packaged as an equivalence of full labels with inverse action by a^{-1}, and Lean proves that the zero label is fixed exactly. Consequently Lean proves that full-label sums and full-label averages are invariant under this multiplicative permutation; the same holds for sums over the subordinate nonzero full-label subset. These invariance statements are also specialized to the Gaussian degree function $G(a)$, so the full Gaussian average and the subordinate Gaussian sum are invariant under multiplicative reindexing. At the coordinate level, Lean proves the corresponding sharp contrast: unit multiplication preserves the predicate $j \ll 0$, whereas additive translation preserves this predicate for all j if and only if the translation is zero. More strongly, an affine coordinate operation $j \mapsto aj + t$, with $a \in F_\ell^\times$, descends to full absolute labels if and only if $t = 0$, and preserves the subordinate predicate for all j if and only if $t = 0$. Lean separately proves that raw F_ℓ-coordinate sums are invariant under additive translation, unit multiplication, and their affine composition; this is only finite-permutation invariance and does not supply descent to $|F_\ell|$. For the Gaussian coordinate average, Lean packages this as a diagnostic: every affine map with nonzero additive offset preserves the raw coordinate average but fails full-label descent. If the Gaussian environment degree is nonzero, the same affine operation also changes the zero Gaussian component, so aggregate average invariance is not pointwise Gaussian compatibility. Lean also proves the exact degeneracy criterion: a coordinate Gaussian component is zero if and only if its coordinate is zero or the environment degree is zero; for an affine zero-coordinate image this becomes $t = 0$ or $\deg(q) = 0$. At the purely labelwise level, Lean proves the exact pointwise preservation criterion: $j \mapsto aj + t$ preserves the full absolute label of every coordinate if and only if $t = 0$ and $a \in \{\pm 1\}$. At the Gaussian-value level, the sign subgroup with $t = 0$ preserves coordinate Gaussian degrees pointwise; conversely, when the environment degree is nonzero, any affine pointwise Gaussian preservation already forces $t = 0$. Lean also converts the zero/nonzero cusp-label compatibility object to a uniform F_ℓ average and proves that a diagonal constant log-volume distribution averages to that same constant;

- the IUT III Step (x) averaged-log-volume corridor: Lean records that (Ind1) and (Ind2) preserve the procession-normalized averaged log-volume, while (Ind3) supplies an upper bound;

the before-, after-(Ind1)-, and after-(Ind2)-averages are all bounded by the same (Ind3) upper bound;

- the Step (x) to Step (xi) upper-ray handoff: if the q-pilot reading is the pre-indeterminacy averaged log-volume and the Θ-hull reading is the (Ind3) upper bound, Lean constructs the hull/determinant upper-ray datum used in Step (xi);

- the Step (x) to Step (xi-g) two-computation handoff: the same data constructs the q-pilot two-computation record, with the input prime-strip reading equal to the pre-indeterminacy average and bounded by the Θ-hull upper-ray value;

- the Step (x)-sourced C_Θ endpoint: adding $|\log(q)| > 0$ for the pre-indeterminacy q-pilot average and the displayed upper bound $-|\log(\Theta)| \leq C_\Theta |\log(q)|$ yields $C_\Theta \geq -1$;

- the Step (x)-sourced statement endpoint: after adjoining the statement-level Θ-subject/q-not-subject boundary, the same data constructs the Corollary 3.12 statement endpoint, with finite Θ-branch excluding $+\infty$ and $C_\Theta \geq -1$; Lean proves that this constructed endpoint still records the Θ-pilot as subject to (Ind1), (Ind2), (Ind3) and the q-pilot as not subject to them, and that these two status fields are unequal at the constructed endpoint; Lean also proves that the endpoint q-pilot log-volume is exactly the pre-indeterminacy average and, by Step (x) invariance, also the after-(Ind1)- and after-(Ind2)-averages; thus $|\log(q)|$ is the negative of these equal averages and is positive; equivalently, the three equal averages are negative under the q-pilot positivity hypothesis; Lean also proves the displayed inequality $-|\log(q)| \leq -|\log(\Theta)|$ at this endpoint, with q read as the pre-indeterminacy average and Θ read as the finite real branch; equivalently, the endpoint q-pilot log-volume lies in the constructed upper ray $\mathbb{R}_{\leq -|\log(\Theta)|}$; Lean also formalizes the opening reduction in the printed proof: if this finite real Θ-value is nonnegative, then the strict negativity of the q-pilot log-volume gives the comparison immediately; moreover the displayed bound $-|\log(\Theta)| \leq C_\Theta |\log(q)|$ then forces $C_\Theta \geq 0$; the finite extended Θ-value and the real Θ-branch are identified with the Step (x) (Ind3) upper bound; the same endpoint bound holds after the (Ind1) and (Ind2) average-preserving transformations; the Step (x)-sourced endpoint also directly rejects $C_\Theta < -1$;

- base-valuation products of tensor packets, the finite $(F^\times_{\mathrm{mod}})_j$-action shadow, and a coric transfer shadow $F \to D \to D^\vdash$ preserving product log-volume without identifying Hodge-theater histories;

- the Step (xi-d)–(xi-g) shadow: determinant normalization, upper-ray membership, and equality of the two q-pilot log-volume computations;

- the Corollary 3.12 finiteness endpoint: the statement-level value $-|\log(\Theta)| \in \mathbb{R} \cup \{+\infty\}$ is represented by an explicit finite/$+\infty$ split, and the hull/determinant upper-ray datum forces the finite real branch, explicitly excluding the $+\infty$ branch; Lean also exposes the resulting real Θ-log-volume and proves the q-pilot upper-ray comparison against that real value;

- the Corollary 3.12 statement boundary: the Θ-pilot log-volume is typed as subject to (Ind1), (Ind2), (Ind3), while the q-pilot log-volume is typed as not subject to these indeterminacies;

- the extended statement endpoint: from the finite Θ-branch, $0 < -|\log(q)|$, the upper-ray comparison, and the displayed hypothesis $-|\log(\Theta)| \leq C_\Theta |\log(q)|$, Lean proves $C_\Theta \geq -1$;

- the Step (xi-f) final algebra: from $-|\log(q)| \leq -|\log(\Theta)|$, $|\log(q)| > 0$, and $-|\log(\Theta)| \leq C_\Theta|\log(q)|$, Lean proves $C_\Theta \geq -1$;
- the signed-payload bridge: the existing Stage 1 comparison record $q_{\text{signed}} \leq \Theta_{\text{signed}}$, together with $\Theta_{\text{signed}} \leq C_\Theta(-q_{\text{signed}})$, feeds this Step (xi-f) algebra directly;
- the Step (xi-g) endpoint bridge: the two-computation q-pilot record produces the signed comparison payload with q_{signed} read from the input prime-strip log-volume and Θ_{signed} read from the hull/determinant upper-ray boundary;
- the Step (xi-e)–(xi-f) fixed-value condition: in the signed two-computation endpoint, Lean proves that the input prime-strip reading and the hull-output reading are both the same real value $-|\log(q)|$, and that this value lies in the upper ray $\mathbb{R}_{\leq -|\log(\Theta)|}$;
- the fixed-value C_Θ endpoint: combining the preceding fixed upper-ray membership with $-|\log(\Theta)| \leq C_\Theta|\log(q)|$, Lean derives $-|\log(q)| \leq C_\Theta|\log(q)|$ and hence $C_\Theta \geq -1$; consequently the same hypotheses reject any $C_\Theta < -1$;
- the statement-endpoint composition: the two-computation C_Θ endpoint canonically supplies the finite Θ-branch Corollary 3.12 statement endpoint once the Θ-subject/q-not-subject indeterminacy boundary is provided;
- the opening sign reduction in the proof of Corollary 3.12: from $0 < -q_{\text{signed}}$, Lean proves that $0 \leq \Theta_{\text{signed}}$ immediately implies $q_{\text{signed}} \leq \Theta_{\text{signed}}$, matching the printed reduction to the case $-|\log(\Theta)| < 0$; the same strict-negativity argument is now available at the statement endpoint for the real Θ-log-volume;
- the Step (xi-h) warning: for $N \geq 2$ and $\Theta < 0$, the tensor-power boundary satisfies $N\Theta < \Theta$;
- the Step (xii) shadow: local Frobenioid p_v^N-ambiguity changes log-volume by an integer shift, while global calibration fixes exponent 0;
- the Remark 3.12.1(iii) arithmetic fact $\mathbb{Q}_{>0} \cap \mathbb{Z}^\times = \{1\}$;
- the Remark 3.12.1(ii) generalized-link numerical bound $C_\Theta \geq -\lambda$ for $\lambda \in \mathbb{Q}_{>0}$, with strict strengthening over the $\lambda = 1$ bound when $\lambda < 1$;
- the q^λ-pilot endpoint algebra: from a signed comparison for $-\lambda|\log(q)|$ and the same $C_\Theta|\log(q)|$ upper bound, Lean derives $C_\Theta \geq -\lambda$, and hence $C_\Theta \geq -1$ when $\lambda \leq 1$;
- the Remark 3.12.2 distinct-label implication q-itw $\Rightarrow$ q-itw $\wedge$ Θ-itw/indets, with q-native and Θ-native labels proved distinct;
- the Remark 3.12.2 toy calculation: forgotten concrete assignments $-h$ and $-2h$ are incompatible for $h > 0$, while $-h \in \mathbb{R}_{\leq -2h+\epsilon}$ implies $h \leq \epsilon$;
- the Fig. 3.9 column table: both $\bullet_0$ and $\circ$ roles are required in both columns, while q-pilot multiradiality is absent;
- the Remark 3.12.2(v) absorption distinction: zero-column unit indeterminacy is absorbed by a hull upper ray, while one-column logarithmic conjugacy is absorbed as log-volume equality;
- the Remark 3.12.3(i) metric analogy: $\mathrm{vol}(S) > 0$ and $\chi_S = -\mathrm{vol}(S)$ imply $\chi_S < 0$;

- the Remark 3.12.4(iii) factor-p analogy: theta-label normalization divides a gap log-volume by ℓ, and multiplication by ℓ recovers it;
- the Remark 3.12.4(v) p-adic analogue: an inclusion, not isomorphism, gives $(1-p)(2g-2) \leq 0$;
- the IUT IV, Theorem 1.10 algebraic handoff: an explicit expression for C_Θ, together with $C_\Theta \geq -1$ and the remaining elementary estimates, implies the displayed $\frac{1}{6}\log(q)$ bound;
- the final elementary coefficient estimates in IUT IV, Theorem 1.10: $(1-12/\ell^2)^{-1} \leq 2$ and $(1-12/\ell^2)^{-1}\frac{1}{4}(1+36d_{\mathrm{mod}}/\ell) \leq 1+80d_{\mathrm{mod}}/\ell$;
- the IUT IV, Theorem 1.10 base-change monotonicity: $\log(d_{F_{\mathrm{tpd}}}) + \log(f_{F_{\mathrm{tpd}}}) \leq \log(d_F) + \log(f_F)$ propagates through the final right-hand side;
- the IUT IV, Theorem 1.10 Step (i) identities: $n^{-1}\sum_{m=1}^{n} m = (n+1)/2$ and $n^{-1}\sum_{m=1}^{n} m^2 = (2n+1)(n+1)/6$;
- the IUT IV, Theorem 1.10 Step (ii) small-prime error: $\log(2^{11}3^3 5^2) \leq 21$, represented by the corresponding linear logarithmic bound;
- the IUT IV, Theorem 1.10 Step (i) GL_2 arithmetic constants for $p = 2, 3, 5$, including the product with the quadratic factor from (E5);
- the IUT IV, Theorem 1.10 Step (iii) algebraic estimate combining the Step (ii) K-bound and the displayed $\log(s_Q)$-bound into $(1+4/\ell)\log(d_K, f_K) + (16/\ell)\log(s_Q) \leq (1+36d_{\mathrm{mod}}/\ell)S + 2\log(\ell) + 70$;
- the IUT IV, Theorem 1.10 Step (viii) Proposition 1.6 case split: the two cases $e^*_{\mathrm{mod}}\ell \geq \eta_{\mathrm{prm}}$ and $e^*_{\mathrm{mod}}\ell < \eta_{\mathrm{prm}}$ imply the uniform $4/3(e^*_{\mathrm{mod}}\ell + \eta_{\mathrm{prm}})$ bound;
- the IUT IV, Theorem 1.10 Step (viii) final absorption of $2\log(\ell) + 74$ and the prime-product term into $10(e^*_{\mathrm{mod}}\ell + \eta_{\mathrm{prm}})$;
- the IUT IV, Theorem 1.10 final display: the Corollary 3.12 C_Θ handoff, followed by the tripodal-to-F monotonicity, gives the final F-level $\frac{1}{6}\log(q)$ inequality;
- the IUT IV, Theorem A bounded-discrepancy conclusion: boundedness below of $(1+\epsilon)(\log \mathrm{diff}_X + \log \mathrm{cond}_D) - \mathrm{ht}_{\omega_X(D)}$ implies the corresponding pointwise height inequality on points of bounded degree;
- the IUT IV, Corollary 2.2(i) bounded-discrepancy chain: $\frac{1}{6}\log(q_2) \approx \frac{1}{6}\log(q_\forall) \approx \frac{1}{6}\mathrm{ht}_\infty \approx \mathrm{ht}_{\omega_X(D)}$ composes to a direct bounded discrepancy relation;
- bounded-discrepancy transfer lemmas: pointwise upper and lower bounds move across an $\approx$ relation with the expected additive constant shift;
- the IUT IV, Corollary 2.2(ii) condition (C1) prime-scale window: $(\log(q_\forall))^{1/2} \leq \ell \leq 10\delta(\log(q_\forall))^{1/2}\log(2\delta\log(q_\forall))$. Lean keeps the square-root and logarithmic factor explicit, proves $\log(q_\forall) \geq 0$, and derives $1 \leq 10\delta\log(2\delta\log(q_\forall))$ from the nonempty window;
- the IUT IV, Corollary 2.2(ii) use of (C1) in Theorem 1.10: $\sqrt{h} \leq \ell$ and $80d_{\mathrm{mod}} \leq \delta$ imply $80d_{\mathrm{mod}}/\ell \leq \delta h^{-1/2}$;

- the IUT IV, Corollary 2.2(ii) error-term use of (C1): $d^*_{\mathrm{mod}} \le \delta$ and $\ell \le 10\delta h^{1/2}\log(2\delta h)$ imply $20(d^*_{\mathrm{mod}}\ell + \eta_{\mathrm{prm}}) \le 200\delta^2 h^{1/2}\log(2\delta h) + 20\eta_{\mathrm{prm}}$;

- the IUT IV, Theorem 1.10 to Corollary 2.2(ii) first-bound composition: these two C1 replacements convert the Theorem 1.10 right-hand side into $(1+\delta h^{-1/2})S + 200\delta^2 h^{1/2}\log(2\delta h) + 20\eta_{\mathrm{prm}}$;

- the IUT IV, Corollary 2.2(ii) q_2/q comparison: the displayed estimate $\frac{1}{6}\log(q_2) - \frac{1}{6}\log(q) \le \frac{1}{3}h^{1/2}\log(2\delta h)$ follows through the intermediate term $h^{1/2}\log(\ell)$;

- the IUT IV, Corollary 2.2(ii) pre-ϵ_E bound for $h = \log(q_\forall)$: the Theorem 1.10 bound, the q_2/q estimate, and the bounded discrepancy B_K imply

$$\frac{1}{6}h \le (1+\delta h^{-1/2})S + (15\delta)^2 h^{1/2}\log(2\delta h) + \frac{1}{2}C_K,$$

  with $C_K = 40\eta_{\mathrm{prm}} + 2B_K$;

- the IUT IV, Corollary 2.2(ii) condition (C2) inequality chain: $\frac{1}{6}\log(q) \le \frac{1}{6}\log(q_2) \le \frac{1}{6}\log(q_\forall) \le (1+\epsilon_E)(\log \mathrm{diff}_X + \log \mathrm{cond}_D) + C_K$;

- the IUT IV, Corollary 2.2(ii) definition of ϵ_E: $\epsilon_E = (60\delta)^2 h^{-1/2}\log(2\delta h)$, represented with $h^{1/2}$ and $\log(2\delta h)$ as explicit real parameters; Lean proves positivity and $\epsilon_E \ge 5\delta h^{-1/2}$ from $\delta \ge 2$, $h^{1/2} > 0$, and $\log(2\delta h) \ge 1$;

- the IUT IV, Corollary 2.2(ii) ϵ_E-error conversion: substituting the definition of ϵ_E, Lean proves $(15\delta)^2 h^{1/2}\log(2\delta h) \le \frac{1}{6}h \cdot \frac{2}{5}\epsilon_E$;

- the IUT IV, Corollary 2.2(ii) bridge to the denominator step: the pre-ϵ_E h-bound, the error conversion, and $\delta h^{-1/2} \le \epsilon_E/5$ yield the preliminary inequality used to move the $\frac{2}{5}\epsilon_E\frac{1}{6}h$ term to the left;

- the IUT IV, Corollary 2.2(ii) denominator step: Lean moves the term $(2\epsilon_E/5)\frac{1}{6}h$ to the left and obtains the denominator $(1-2\epsilon_E/5)^{-1}$;

- the IUT IV, Corollary 2.2(ii) final ϵ_E-absorption: from $0 < \epsilon_E \le 1$, Lean proves

$$\frac{1+\epsilon_E/5}{1-2\epsilon_E/5} \le 1+\epsilon_E, \qquad (1-2\epsilon_E/5)^{-1}C_K/2 \le C_K,$$

  and hence the displayed C2 bound from the preceding denominator expression;

- the IUT IV, Corollary 2.2(ii) final h-bound: Lean composes the pre-ϵ_E bound, error conversion, move-left step, and absorption to prove $\frac{1}{6}h \le (1+\epsilon_E)S + C_K$;

- the IUT IV, Corollary 2.2(ii) final C2 bridge: using $S \le \log \mathrm{diff}_X(x_E) + \log \mathrm{cond}_D(x_E)$, Lean derives the displayed C2 inequality chain;

- the IUT IV, Corollary 2.2(ii) comparison $\log \mathrm{diff}_X(x_E) = \log(d_{F_{\mathrm{tpd}}})$ and $\log(f_{F_{\mathrm{tpd}}}) \le \log \mathrm{cond}_D(x_E) \le \log(f_{F_{\mathrm{tpd}}}) + \log(2\ell)$;

- the IUT IV, Corollary 2.2 to Theorem A passage: bounded-discrepancy equivalence $\frac{1}{6}\log(q_2) \approx \mathrm{ht}_{\omega_X(D)}$, together with the C2 bound for $\frac{1}{6}\log(q_2)$, gives a lower bound for $(1+\epsilon_E)(\log \mathrm{diff}_X + \log \mathrm{cond}_D) - \mathrm{ht}_{\omega_X(D)}$;

- the IUT IV, Corollary 2.2/2.3 finite-exception passage: a lower bound on the complement of a finite exceptional set and a lower bound on the exceptional set glue to a global lower bound by taking the minimum of the two constants;

- the IUT IV, Corollary 2.3 Diophantine inequality: Lean records the GenEll transfer as an explicit hypothesis and derives the global bounded-discrepancy lower bound for $(1+\epsilon)(\log \operatorname{diff}_X + \log \operatorname{cond}_D) - \operatorname{ht}_{\omega_X(D)}$;

- comparison levels: pointwise representative, aggregate representative, balanced sign-compatible, and hull/log-volume;

- endpoint audits for the $3.11.5 \Rightarrow 3.12$ comparison chart;

- nonarchimedean and archimedean (Ind3) local orientations;

- ordered real-line alignment with cancellation only after scale equality;

- a canonical unit-scale ordered alignment from a nonarchimedean (Ind3) entry;

- the local-object Hodge-descent square-weight boundary: transported zero-column and cusp-class local objects have the same finite log-volume as the packet object, while square/full-label preservation obligations remain explicit;

- the square/full-label Hodge/SHE transport audit: a pointwise upper bound on the transported target full-label log-volumes transfers to the source square-weighted average once full-label log-volume, square weight, and total-weight preservation are supplied; the same transfer is now available inside the structured-SHE wrapper, and the resulting upper bound is exposed for the square-weighted average in the F_ℓ cusp-label route after matching the source profile and source log-volume; consequently, structured-SHE target full-label bounds by the Θ-source average feed the cusp-class container route all the way to $q_{\text{signed}} \leq \Theta_{\text{signed}}$; the same final comparison is now available directly from the primitive factored square/full-label SHE obligations, without requiring a prebuilt transport-audit argument;

- a Gaussian-to-factored-SHE bridge: source and target degree-level Gaussian evaluations with equal environment degree supply the full-label value-preservation branch of the factored square/full-label SHE obligations; with separate coordinate-square and full-label-map preservation, the experiment route reaches $q_{\text{signed}} \leq \Theta_{\text{signed}}$ directly from Gaussian target bounds; since the absolute-label exponent is nonnegative, Lean also derives those target bounds in the branch where the target environment degree is nonpositive and the Θ-source average is nonnegative; conversely, the all-label Gaussian target-bound formulation forces the Θ-source average to be nonnegative, since the core label has Gaussian degree 0; after removing the core label, Lean proves that every nonzero Gaussian target value is bounded by the Θ-source average if the target environment degree is nonpositive and is itself bounded by that average; Lean also forms the corresponding finite average over the nonzero $ZMod$ carrier, proves the same upper bound for this nonzero average, and proves the exact rescaling between this auxiliary nonzero average and the uniform F_ℓ-average with zero Gaussian term; because the square-weighted route assigns weight 0 to the core label, Lean now feeds nonzero-only target bounds through the square/full-label transport audit all the way to $q_{\text{signed}} \leq \Theta_{\text{signed}}$; this nonzero Gaussian environment-comparison route is now a source-layer theorem, not only an experiment wrapper;

- a bridge from factored square/full-label SHE obligations to the Hodge-descent (Ind3) route.

5 Current Lean Results

The current first-pass experiment report evaluates to:

$$\begin{aligned}
&\text{route theorem flags} &&= \texttt{true},\\
&\text{collapse-obstruction flags} &&= \texttt{true},\\
&\text{Gaussian/procession finite-model flags} &&= \texttt{true},\\
&\texttt{balancedLevelRejectedAtFinalRouteTheoremAvailable} &&= \texttt{true},\\
&\texttt{disputeSettledByCurrentStage} &&= \texttt{false}.
\end{aligned}$$

Here the route flags are ordered real-line, nonarchimedean entry, and factored SHE availability. The obstruction flags are raw-cancellation failure, label-independent j^2 failure, and sign-quotient representative descent failure. The Gaussian/procession flags include finite tensor-packet, monoid action, normalized splitting-generator average, the explicit square-sum formula for the generator total and normalized delta, the resulting conditional monotonicity of the acted tensor packet, canonical positive square-weight profile, the constant-average check, base-valuation product, coric-transfer, determinant-normalization, upper-ray, q-pilot two-computation, the Step (x) averaged-log-volume indeterminacy corridor, the refined Step (x) bounds for the before-, after-(Ind1)-, and after-(Ind2)-averages, tensor-power warning, and global Frobenioid-calibration shadows, plus the finiteness endpoint for $-|\log(\Theta)| \in \mathbb{R} \cup \{+\infty\}$, including extraction of the finite real Θ-value, the direct Corollary 3.12 statement endpoint with Θ-subject/q-not-subject pilot bookkeeping, the Step (x)-sourced upper-ray, q-pilot two-computation, statement endpoint, propagation of the same pilot-status boundary to that endpoint, proof that the endpoint Θ- and q-statuses are distinct, identification of its q-pilot real with the Step (x) pre-, after-(Ind1)-, and after-(Ind2)-averages, positivity of $|\log(q)|$ at that endpoint, and direct $-|\log(q)| \in \mathbb{R}_{\leq -|\log(\Theta)|}$, $-|\log(q)| \leq -|\log(\Theta)|$, finite-$\Theta$ exclusion of $+\infty$, identification of the finite extended Θ-value with the Step (x) (Ind3) upper bound, the nonnegative-Θ opening reduction including the stronger $C_\Theta \geq 0$ branch, and $C_\Theta \not< -1$ conclusions, the Step (xi-g)-to-(xi-f) endpoint bridge giving $C_\Theta \geq -1$ from the two-computation q-pilot record, the Step (xi-e)–(xi-f) fixed $-|\log(q)|$ membership check and its direct $C_\Theta \geq -1$ consequence, the composition into the statement-level endpoint, and positive-rational unit rigidity. The generalized-link flag records the rational parameter inequality $C_\Theta \geq -\lambda$, now with the corresponding q^λ-pilot real-valued derivation. The distinct-label flag records the logical intertwining display in Remark 3.12.2 and rejects collapse of the q-native and Θ-native labels. The toy-model flag records the resulting upper-ray arithmetic $h \leq \epsilon$. The column-table flag records the Fig. 3.9 theta/q similarity-difference split. The absorption flag records the zero-column upper-bound versus one-column equality distinction. The metric flag records the Gauss–Bonnet sign calculation used as analogy for the upper semi-compatibility inequality. The factor-p flag records the F_ℓ-label averaging denominator. The Frobenius-derivative flag records the degree inequality from inclusion. The IUT IV flag records the downstream Theorem 1.10 use of the Corollary 3.12 lower bound on C_Θ, including the final coefficient estimates and the tripodal-to-F log-degree monotonicity. The Step (i) flag records the two elementary sum identities used in the leading-term computation. The Step (ii) small-prime flag records the additive error bound 21. The GL_2 flag records the finite cardinality constants feeding that bound. The Step (iii) flag records the $\log(s_Q)$ coefficient absorption. The Step (viii) flag records the prime-product case split used before the final upper bound for C_Θ, plus the final coarse error absorption. The final display flag records the composed Theorem 1.10 inequality with F-level log-degree terms. The Theorem A flag records the bounded-discrepancy height/log-different/log-conductor inequality. The Corollary 2.2 flag records composition of the displayed bounded-discrepancy chain. The transfer flag records movement of pointwise bounds across bounded-discrepancy equivalence. The Corollary 2.2(C1) flag records the displayed prime-scale window and its elementary consequences. The Corollary 2.2(C2) flag records the displayed three-step logarithmic inequality chain. The C1 coefficient flag records the replacement of $80d_{\mathrm{mod}}/\ell$ by $\delta h^{-1/2}$ in the Theorem 1.10-to-C2 proof. The C1 error-term flag records the corresponding replacement of $20d^*_{\mathrm{mod}}\ell$ by $200\delta^2 h^{1/2}\log(2\delta h)$. The first-bound flag records the composition of those replacements with Theorem 1.10. The ϵ_E-definition flag records the source formula and lower estimate for ϵ_E. The ϵ_E-move-left flag records introduction of the denominator $(1-2\epsilon_E/5)^{-1}$. The ϵ_E-absorption flag records the final denominator removal in C2. The ϵ_E-error flag records the conversion of the $(15\delta)^2$-term to the $\frac{2}{5}\epsilon_E\frac{1}{6}h$ term. The final h-bound flag records the full composition through absorption. The final C2 bridge flag records the last monotonic replacement of the tripodal log-degree sum by the curve different/conductor sum. The h-to-move-left flag records the bridge from the pre-ϵ_E bound to the denominator-stage preliminary inequality. q2/q flag records the displayed comparison of $\log(q_2)$ and $\log(q)$. The pre-ϵ_E h-bound flag records the subsequent absorption of the bounded $q_\forall/q_2$ discrepancy into C_K. The log-different/log-conductor flag records the bridge from tripodal field terms to curve functions. The Corollary 2.2-to-Theorem A flag records the formal bounded-discrepancy transfer from C2 to the Theorem A inequality. The finite-exception flags record the lower-bound gluing used when finite exceptional sets are enlarged in Corollary 2.2 and then passed toward Corollary 2.3. The Corollary 2.3 flag records the final GenEll-transfer interface for the Diophantine inequality. The theorem-level route is:

$$\begin{array}{ccccc}
\text{nonarch. (Ind3) entry} & \to & \text{canonical ordered } \mathbb{R}\text{-alignment} & \to & \text{source/target alignment} \\
 & & \downarrow & & \downarrow \\
\text{factored SHE transport} & \to & \text{Hodge-descent audit} & \to & q_{\mathrm{signed}} \le \Theta_{\mathrm{signed}}.
\end{array}$$

The route remains conditional on the displayed hypotheses. The local-object Hodge-descent layer now exposes equality of the transported finite log-volumes before square/full-label preservation is supplied. The square/full-label audit now also transports target pointwise upper bounds back to the source square-weighted average, including through the structured-SHE route wrapper, and the experiment layer exposes the resulting final q-to-Θ comparison theorem. The same experiment-level theorem is now exposed for primitive factored square/full-label SHE obligations and for degree-level Gaussian evaluations after the explicit preservation branches are supplied. In the nonpositive-environment/nonnegative-Θ-average branch, the Gaussian pointwise target bound is derived from the sign of the q^{j^2}-exponent rather than assumed. Lean also proves that this all-label Gaussian bound cannot hold when the Θ-source average is negative. For nonzero labels, Lean proves the complementary bound: the absolute exponent is at least 1, so a nonpositive environment degree below the Θ-source average bounds every nonzero Gaussian component and hence their finite nonzero-label average. The square-weighted transport route has also been strengthened from an all-label pointwise bound to a weighted-summand bound, so the zero label requires no target upper bound. The experiment theorem now delegates to the source-layer q-to-Θ theorem for this nonzero Gaussian environment-comparison route. Separately, Lean proves that the uniform F_ℓ-average of the Gaussian shadow is $(\ell-1)/\ell$ times the auxiliary nonzero-carrier average, fixing the denominator at ℓ when the zero label is present.

6 Audit Findings

No formal statement currently proves IUT III, Corollary 3.12 from Mochizuki's published hypotheses. The strongest positive result is conditional: if the hull/log-volume upper-ray comparison supplies $-|\log(q)| \le -|\log(\Theta)|$, then Lean checks the formal Step (xi-f) conclusion $C_\Theta \ge -1$. If the nonarchimedean (Ind3) entry, ordered real-line alignment, and Hodge/SHE square-weight transport audit are supplied, Lean checks the final signed q-to-Θ inequality feeding that algebra. If the corresponding primitive factored square/full-label SHE obligations are supplied instead, Lean first constructs the transport audit and checks the same final inequality. If the input is a pair of degree-level Gaussian evaluations with equal environment degree, Lean constructs the full-label log-volume compatibility branch before invoking the factored route. The branch with nonnegative Θ-source average is formally easier: nonnegativity of j^2 makes every nonpositive environment-degree Gaussian target value bounded above by that average. The all-label version cannot be the negative branch: the core label contributes 0, so a negative upper bound is impossible. On the nonzero-label part, the formal obstruction disappears: exponent ≥ 1 gives the expected bound from the environment comparison. The square-weighted final route now accepts this nonzero-only bound, since the core summand has weight 0. This gives a conditional source-level q-to-Θ route from degree-level Gaussian data, coordinate-square preservation, full-label-map preservation, equal source/target environment degree, and environment degree bounded above by the Θ-source average. The zero/nonzero average split is now exact: the nonzero carrier has cardinality $\ell - 1 = 2j_{\max}$, the signed carrier is a twofold cover of the nonzero absolute half-range, the zero Gaussian degree is 0, and the all-F_ℓ average is $(\ell-1)/\ell$ times the nonzero absolute average; consequently the F_ℓ-coordinate and full $|F_\ell|$-absolute averages differ strictly for nonzero environment degree. The remaining task is to replace the current degree-level Gaussian shadow by the actual Hodge–Arakelov Gaussian/Frobenioid construction behind this environment comparison.

The strongest negative result is also limited: in the current $\mathbb{F}_\ell$-label model, Lean rejects a single label-independent scale accounting for all representative j^2 factors, and it rejects balanced sign-compatible evidence as the final hull/log-volume comparison level. Lean also rejects direct descent of arbitrary raw representative j^2 theta-degree data to $|F_\ell|$, while accepting the half-range exponent convention stated in IUT II. Lean also checks that an N-th tensor-power boundary is a strict strengthening for $N \geq 2$ and negative pilot log-volume. It also checks the Step (xii) numerical warning: nonzero local Frobenioid shifts change log-volume, whereas global calibration restores the unshifted value. The positive rational unit rigidity fact in Remark 3.12.1(iii) is formalized directly over $\mathbb{Q}$. For generalized links, Lean records that $\lambda < 1$ makes $-\lambda > -1$. Lean also checks that the Remark 3.12.2 simultaneous-intertwining step depends on distinct labels rather than an unlabeled identification. In the accompanying toy model, Lean proves the final upper bound exactly from the upper-ray membership hypothesis. Lean also records that the q-pilot column lacks the theta-side multiradiality used in Theorem 3.11(iii)(c). For Remark 3.12.2(v), Lean separates hull absorption of unit shifts from log-volume equality after logarithmic-conjugate compatibility. For Remark 3.12.3(i), Lean proves the stated negative Euler-characteristic sign from positivity of metric volume. For Remark 3.12.4(iii), Lean checks the ℓ-denominator normalization identity for the theta-label averaging analogy. For Remark 3.12.4(v), Lean proves the displayed nonpositive degree defect from $p \geq 2$ and $g \geq 2$. For IUT IV, Theorem 1.10, Lean proves the algebraic implication from the computed C_Θ expression and $C_\Theta \geq -1$ to the final $\frac{1}{6}\log(q)$ estimate, conditional on the separate elementary estimates. Lean now also proves the two displayed final elementary coefficient estimates from $\ell \geq 5$. It also proves the final monotonic replacement of the tripodal log-degree sum by the larger F-level log-degree sum, conditional on the two component monotonicity inequalities cited from Proposition 1.3(i). Lean also proves the Step (i) sum and square-sum identities over $\mathbb{R}$. For Step (ii), Lean proves the small-prime additive error bound from the displayed logarithmic inequalities. Lean also checks the Step (i) GL_2 constants for $p = 2, 3, 5$ and their product with the quadratic extension factor. For Step (iii), Lean proves the displayed coefficient absorption from the source hypotheses $d_{\mathrm{mod}} \geq 1$, $\ell \geq 5$, $2\log(\ell) \leq \ell$, and nonnegativity of the tripodal log-degree sum. For Step (viii), Lean checks that the two Proposition 1.6 cases imply the uniform prime-product bound used in the final C_Θ estimate. Lean also checks the subsequent absorption into the coefficient 10 of $e^*_{\mathrm{mod}}\ell + \eta_{\mathrm{prm}}$. Combining the Corollary 3.12 handoff with the tripodal-to-

F monotonicity, Lean proves the final displayed F-level inequality of Theorem 1.10 at this real-valued shadow level. For Theorem A, Lean records the exact bounded-discrepancy conclusion and proves the equivalent pointwise height bound from the lower-bound hypothesis. For Corollary 2.2(i), Lean defines equality up to bounded discrepancy, proves reflexivity, symmetry, transitivity, nonnegative scaling, and composes the displayed chain from $\log(q_2)$ to $\mathrm{ht}_{\omega_X(D)}$. Lean also proves that upper and lower pointwise bounds transfer across such a bounded-discrepancy equivalence with the appropriate constant shift. For Corollary 2.2(ii)(C1), Lean records the displayed window for ℓ and derives nonnegativity of $\log(q_\forall)$, positivity of $(\log(q_\forall))^{1/2}$, and $1 \le 10\delta\log(2\delta\log(q_\forall))$. Lean also verifies the C1 coefficient replacement $80d_{\mathrm{mod}}/\ell \le \delta h^{-1/2}$ used immediately after applying Theorem 1.10. Lean also verifies the C1 upper-window replacement for the Theorem 1.10 error term $20(d^*_{\mathrm{mod}}\ell + \eta_{\mathrm{prm}})$. Combining these, Lean proves the first displayed post-Theorem-1.10 bound in the proof of Corollary 2.2(ii). Lean also proves the displayed comparison $\frac{1}{6}\log(q_2) - \frac{1}{6}\log(q) \le \frac{1}{3}h^{1/2}\log(2\delta h)$ from the intermediate $h^{1/2}\log(\ell)$ bounds. Lean then combines this with the bounded $q_\forall/q_2$ discrepancy to prove the displayed pre-ϵ_E bound for $\frac{1}{6}h$. For Corollary 2.2(ii), Lean proves the final $\frac{1}{6}\log(q)$ bound from the displayed $\log(q)$, $\log(q_2)$, and $\log(q_\forall)$ chain. Lean records the proof's definition of ϵ_E and verifies the stated lower estimate $\epsilon_E \ge 5\delta h^{-1/2}$. Lean also verifies the source conversion $(15\delta)^2h^{1/2}\log(2\delta h) \le \frac{1}{6}h \cdot \frac{2}{5}\epsilon_E$. Combining this with $\delta h^{-1/2} \le \epsilon_E/5$, Lean proves the preliminary inequality needed for the move-left step. Lean then verifies the algebraic move-left step that introduces the denominator $(1 - 2\epsilon_E/5)^{-1}$. Lean also proves the final ϵ_E-absorption step used just before C2: the intermediate expression with denominator $1 - 2\epsilon_E/5$ is bounded by $(1 + \epsilon_E)$ times the log-degree sum plus C_K. Lean now composes these steps into the final displayed $h = \log(q_\forall)$ bound. Lean also derives the displayed C2 chain from this h-bound and the tripodal-to-curve log-degree comparison. It also proves the two sum inequalities relating the tripodal different and conductor terms to $\log\mathrm{diff}_X + \log\mathrm{cond}_D$. For the Corollary 2.2/2.3 exceptional-set passage, Lean proves that a lower bound off a finite exceptional set and a lower bound on the exceptional set imply a global lower bound, and specializes this to the Theorem A discrepancy. For Corollary 2.3, Lean proves the stated bounded-discrepancy conclusion from an explicit GenEll transfer hypothesis; GenEll itself is not formalized here. Finally, Lean combines the Corollary 2.2(i) bounded-discrepancy height comparison with the C2 bound to construct the Theorem A lower-bound shadow.

7 Missing Mathematics

The present code does not yet formalize:

- initial Θ-data as in IUT I;
- actual Hodge theaters, Frobenioids, prime strips, log-shells, or Θ-links;
- the Hodge–Arakelov theta evaluation and Gaussian monoids of IUT II;
- the construction of the actual theta values, Gaussian/splitting monoids, $F^{\times}_{\mathrm{mod}}$-actions, Kummer isomorphisms, mono-analyticization, and Frobenioids behind the present shadows;
- the actual construction of log-shells and tensor packets attached to mono-analytic prime-strips in IUT III, Theorem 3.11;
- the genuine holomorphic hull and determinant operations of Step (xi), beyond the present finite determinant/upper-ray/two-computation/tensor-power shadows;
- the global realified Frobenioids of Step (xii), beyond the present integer log-volume shift model;

- the arithmetic-divisor and local log-volume construction behind the IUT IV Theorem 1.10 numerical estimates, and the Diophantine applications;
- the construction of the normalized height, log-different, and log-conductor functions used in IUT IV, Theorem A;
- the construction of the bounded-discrepancy equivalences asserted in IUT IV, Corollary 2.2(i);
- a proof that the current Hodge/SHE transport obligations follow from Mochizuki's full hypotheses.

8 Next Work

The next significant formal step is not another wrapper. It is to replace current explicit Hodge/SHE transport assumptions by constructions closer to the papers:

$$\begin{gathered}\text{theta values } q^{j^2}\\ \rightsquigarrow \text{ Gaussian and splitting monoids}\\ \rightsquigarrow \text{ square/full-label preservation}\\ \rightsquigarrow \text{ hull/log-volume route.}\end{gathered}$$

Only after this bridge is formalized can the first-pass experiment begin to test the actual mathematical load-bearing part of Mochizuki's response to Scholze–Stix.

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